Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2023	ENGI3481-WE01

Title: (BEng/MEng)

Power Semiconductor Devices

Time Allowed:	2 hours
Additional Material provided:	Yes, datasheet included as part of exam paper.
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Question 1

Strained semiconductors are materials in which the bond length is different than that in their conventional (unstrained) counterparts. This question deals with strained silicon. In silicon, the bond energy is given by:

$$E_{Si-Si} = -\frac{A}{r} + \frac{B}{r^6}$$

where $A = 0.96$ eV nm and $B = 1.17 \times 10^{-4}$ eV nm^6. The band gap in strained silicon is modulated with the strain following:

$$E_g(\epsilon) = E_g^\circ - \epsilon \cdot \Psi$$

where $E_g^\circ = 1.12$ eV is the band gap energy of an unstrained silicon, ϵ is the unitless strain defined as $\epsilon = (r_{\text{strained}} - r_{\text{eq}})/r_{\text{eq}}$, and $\Psi = 12.3$ eV.

a. Calculate the equilibrium (unstrained) bond length, r° and energy, E°_{Si-Si}. Give your answers in nm and eV, respectively.

[8%]

b. We attempt to create a photodetector for communications wavelengths $\lambda = 1.55$ μm by straining silicon
   i. What is our target (maximal) band gap? Give your answer in eV.
   ii. What is the strain required to achieve this band gap?
   iii. What is the change in bond energy created by this strain? Give your answer in eV. **If you did not solve Sec. a.**, you may use $r^\circ = 0.3$ nm, $E^\circ = -3.1$ eV.

[24%]

Steady state conductivity of two p-type silicon chips, one unstrained ($E_g = 1.12$ eV, $n_i = 10^{10}$ cm$^{-}3$, $\mu_n = 1,400$ cm^{2}V^{-1}s^{-1}, $\mu_p = 450$ cm^{2}V^{-1}s^{-1}) and one strained to produce the material found in Sec. b., is measured under dark conditions and under uniform illumination. Both samples are doped with acceptors density of $N_A = 2 \times 10^{16}$ cm^{-3}, and are illuminated with the same source producing the same generation rate, G_L. The results are given in table **Q1**. The units are $(\Omega^{-1} \cdot \text{cm}^{-1})$.

	Unstrained	Strained
Dark	1.14×10^{-12}	1.35×10^{-7}
Illuminated	6.67×10^{-4}	1.45×10^{-2}

Table Q1

c. Assume low injection, so that $\delta n \ll p$. Calculate the electron mobility in the strained sample, if it is known that the minority lifetime of the strained sample is $\tau_n^{(\epsilon)} = 15 \times \tau_n^\circ$, where τ_n° is the minority lifetime in the unstrained sample.

[40%]

At time $t = T$, the uniform illumination for both samples is turned off.

d. i. Write general expressions for the time-dependent conductivity of excess carriers for each of the samples, using the dark conductivities, $\{\sigma_d^\circ,\ \sigma_d^{(\epsilon)}\}$ as parameters.
   ii. Which sample would reach conductivity saturation (steady-state) value quicker?
   iii. Draw a single graph showing the change in conductivity over time for both samples.

[28%]

Question 2

The made-up semiconductor Durhamnium (chemical symbol Du), has several unknown properties. To characterise it, a piece of Du is doped by donors at a density of $N_D = 3 \times 10^{16}$ cm^{-3}, while a separate piece is doped by acceptors at a density of $N_A = 10^{18}$ cm^{-3}. Both pieces are coated with titanium ($\Phi_M = 4.3$ eV) to produce Schottky contacts.

a. Explain, using appropriate band diagrams, how it is possible that both junctions are rectifying.

[24%]

The forward current densities at a bias of $V_A = 0.3$ V for each of the junctions, at temperatures of 280 and 300 K are given in Table **Q2**. The units are (A cm^{-2}).

Current density, $V_A = 0.3$ V	280 K	300 K
metal–p-type	3.6	10.9
metal–n-type	0.32	1.1

Table Q2

b. i. Calculate the barrier heights, Φ_{Bn} and Φ_{Bp}.

ii. Calculate the magnitude of the band gap of Du.

[44%]

The built-in potential for the two Schottky junctions were found to be $V_{bi}^{(n)} = 0.49$ V and $V_{bi}^{(p)} = 0.58$ V, respectively.

c. i. What would be the built-in potential of a Du p-n junction, if the doping levels were identical to the profiles used for the Schottky junctions?

ii. What is the intrinsic density of carriers, n_i in Du?

[32%]

Question 3

Using the values provided in the datasheet, design a power diode that can be manufactured from Germanium. Assume that the critical electric field of Germanium is 100 kV/cm. The drift region is doped n-type with a concentration that is ten times the intrinsic carrier concentration.

a. Determine the maximum reverse bias voltage that can be applied to the diode if the maximum electric field equals the critical electric field.

[20%]

b. Determine the maximum depletion region width of the diode when operated at the maximum reverse bias.

[16%]

c. Calculate the specific on-state resistance of the diode at 300 K.

[20%]

d. Determine the on-state voltage for the diode operating at a current density of 200 Acm^{-2}.

[16%]

e. Determine the ratio of the capacitance of the diode at zero bias and the maximum reverse bias if it acts like a parallel plane capacitor in a similar manner to a Schottky diode.

[28%]

Question 4

The resistance of a silicon carbide power MOSFET can be expressed as the sum of the resistance of the drift region and the resistance of the channel. The resistance of the channel can be described using the conventional equation for the drain – source current. The MOSFET has an oxide capacitance of 173 nF/cm^2 , a threshold voltage of 12 V and is operated with a gate-source voltage of 20 V.

a. Calculate the aspect ratio (W/L) for the MOSFET, if it is capable of operating at a current of 100 A, with V_{DS}=0.1 V.

[16%]

b. Starting from the conventional equation for the drain – source current, determine the resistance of the channel using the assumption that

$$V_D \ll (V_{GS} - V_T)$$

[32%]

c. The MOSFET has a maximum operating voltage of 1200 V, determine the overall resistance of the device if the drift region dopant concentration is 1×10^{15} cm^{-3} and the device area is 0.4 cm^2.

[20%]

d. Predict the resistance of the MOSFET if it is used in a cryogenic system, where the operating temperature is 77 K.

[32%]

END

[ENGI3331 and ENGI3481]

Datasheet

Quantity	Symbol	Value
Boltzmann Constant	k_B	$1.38066\times10^{-23}\,J/K$
Elementary Charge	e	$1.60218\times10^{-19}\,C$
Electron Rest Mass	m_o	$9.1095\times10^{-31}\,kg$
Electron Volt	eV	$1eV=1.60218\times10^{-19}\,J$
Permeability in vacuum	μ_o	$1.25663\times10^{-8}\,H/cm$
Permittivity in vacuum	ε_o	$8.854\times10^{-14}\,F/cm$
Planck's Constant	h	$6.62617\times10^{-34}\,J\,s$
Reduced Planck's Constant	$\hbar$	$1.05458\times10^{-34}\,J\,s$
Proton Rest Mass	m_p	$1.67264\times10^{-27}\,kg$
Speed of Light in vacuum	c	$2.99792\times10^{10}\,cm/s$
Thermal Voltage at 300K	k_BT/e	$0.0259V$
Wavelength of 1eV Quantum	λ	$1.23977\,\mu m$

Properties of Semiconductor Materials at 300K

Properties	Si	Ge	GaAs	SiC	Diamond	SiO_2
Dielectric Constant	11.9	16	13.1	9.66	5.5	3.9
Effective Density of States in Conduction Band (cm^{-3})	$2.8\text{x}10^{19}$	$1.04\text{x}10^{19}$	$4.7\text{x}10^{17}$	$1.7\text{x}10^{19}$	$1\text{x}10^{20}$	
Effective Density of States in Valance Band (cm^{-3})	$1.04\text{x}10^{19}$	$6.0\text{x}10^{19}$	$7.0\text{x}10^{18}$	$2.5\text{x}10^{19}$	$1\text{x}10^{19}$	
Effective Mass (Electrons)	$m_l=0.98$ $m_t=0.19$	$m_l=1.64$ $m_t=0.082$	$m_l=0.067$	$m_l=0.29$ $m_t=0.42$	$m_l=1.40$ $m_t=0.36$	
Effective Mass (Holes)	$m_l=0.16$ $m_t=0.49$	$m_l=0.044$ $m_t=0.28$	$m_l=0.082$ $m_t=0.45$	$m_l=1.00$ $m_t=1.00$	$m_l=0.80$ $m_t=0.80$	
Energy Gap at 300K (eV)	1.12	0.66	1.424	3.23	5.45	9.0
Critical Electric Field (MV/cm)	0.3		0.4	2.5	10	10
Intrinsic Carrier Conc. (cm-3)	$1.45\text{x}10^{10}$	$2.4\text{x}10^{13}$	$1.79\text{x}10^{6}$	10^{-9}	10^{-27}	
Minority Carrier Lifetime (s)	$2.5\text{x}10^{-3}$	10^{-3}	10^{-8}	$6\text{x}10^{-7}$		
Electron Drift Mobility ($cm^2V^{-1}s^{-1}$)	1500	3900	8500	900	1800	
Hole Drift Mobility ($cm^2V^{-1}s^{-1}$)	450	1900	400	400	1500	
Negative Differential Mob. ($cm^2V^{-1}s^{-1}$)				-100		